

RAD-SMART

The right patient, on the right machine, at a predictable time.

Radiotherapy AI-assisted Dynamic Scheduling, Machine Allocation and Resource Tracking



One planned treatment day: each block is a patient's session (routine, new start, complex, blood irradiation)

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Assistive, not diagnostic

THE PROBLEM

Every weekday, 70–90 patients wait for one machine

Appointment times are given by hand, without knowing how many machine-minutes each patient needs. Too many people are told to come at once, and they wait, every day of a 5–7 week course, usually with a caregiver.

OUR DEPARTMENT'S RECORDS · 55 DAYS, OCT–DEC 2024 · 3,137 PATIENT-DAYS

109 min

one patient in ten waits longer than this, from arrival to treatment

22%

treated more than an hour after their appointment; only 29% within 15 minutes

19 → 66 min

median wait, arriving before 10:00 against after 19:00: delays build all day

The load is rising, from 54 patients a weekday in October to 67 in December, and busy days are worse. 69% of India's 554 radiotherapy facilities have only one machine.

Allocation under hard rules and soft priorities

Hard rules: never broken

- New starts finish by 17:00
- SBRT, SRT, TBI, CSI in 10:00–17:00
- Blood irradiation 13:30–14:00, daily
- 1 breast board, 1 ABC, shared with CT
- MHRC: 16:00–17:00, then the 17:00 bus
- Public transport home by 21:00
- Urgent starts treated by 18:00

Soft priorities: traded off

- Paying patients' preferred times
- Older patients earlier
- Ward, dormitory, nearby: late evening
- Complex cases in 12:00–13:30
- A day of 08:30 to 01:00, little overtime

And every day brings events: machine faults, overruns, no-shows, urgent add-ons and imaging holds, each of which should trigger a re-plan.

FIVE APPROACHES

We compared five ways to build it

Approach	Strength	Weakness	Score
Rule-based slot templates	Simple, transparent	Can't absorb variation or faults	3.10
LLM-only agent	Conversational	No guarantee on hard rules	2.60
Static optimisation	Guarantees hard rules	No learning or live recovery	3.60
Deep reinforcement learning	Adaptive in principle	Opaque; needs huge data	2.45
Hybrid (selected): predict, optimise, simulate, explain	Best of each technique	More parts, built in layers	4.85

Weighted score out of 5: safety and scope 20%, impact 20%, real-time 15%, feasibility 15%, explainability 10%, ease and languages 10%, configurability 10%.

THE SOLUTION

How RAD-SMART works, and who decides

1 · PREDICT

Minutes, not head-counts

Expected and cautious (P80) minutes per session, starting from the department's estimate table.

PoC: 2.1 min error vs 2.6 for averages

2 · OPTIMISE

Hard rules guaranteed

A solver builds the evening-before plan and repairs it live; hard rules cannot break.

PoC: 87 patients, all 7 rules re-checked

3 · SIMULATE

Tested before it is used

A twin calibrated on the department's records tests every plan and forecasts three weeks of load.

PoC: 300 days; 2 machines, peak load 154% → 98%

4 · EXPLAIN

Kannada, Tulu and more

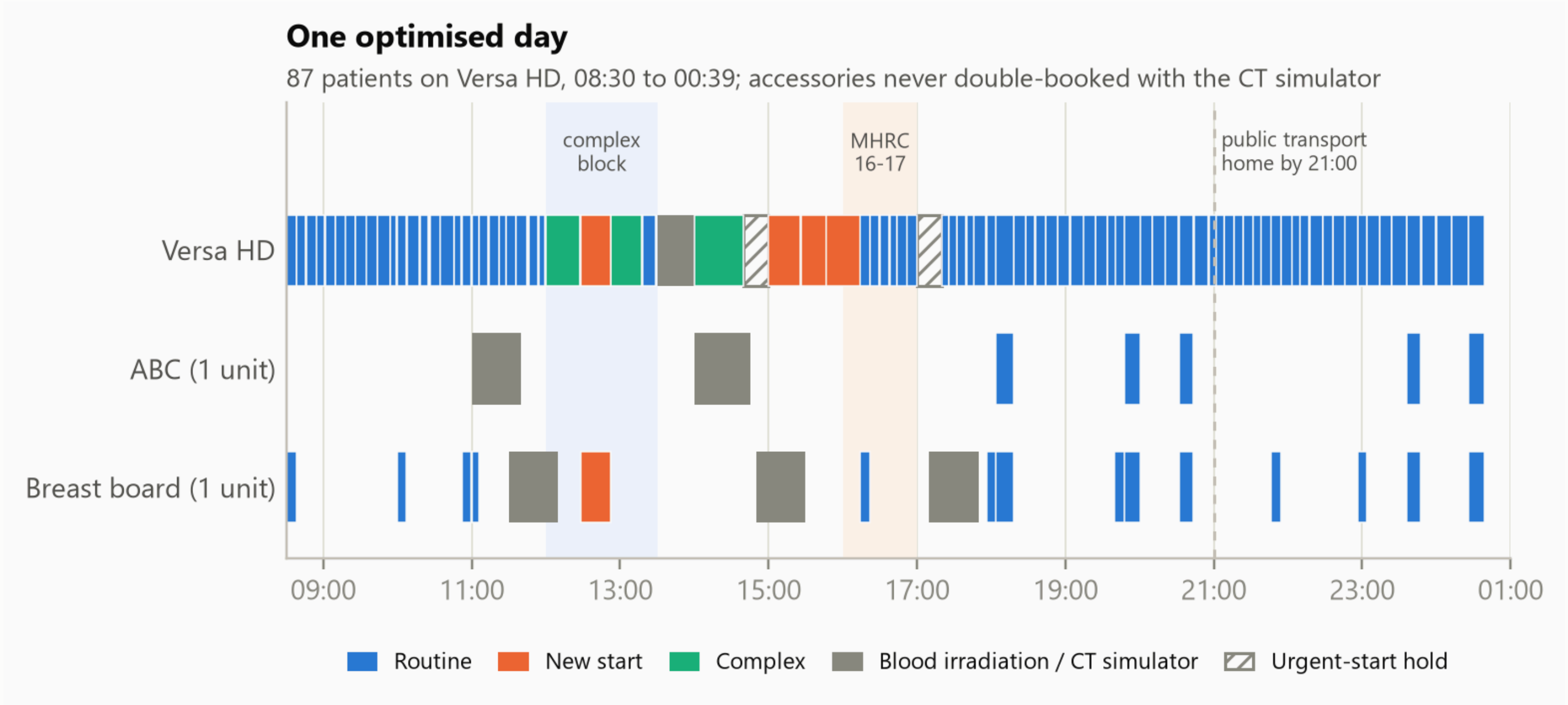
A Sarvam AI copilot explains placements, edits rules from plain language and messages patients.

The LLM never schedules

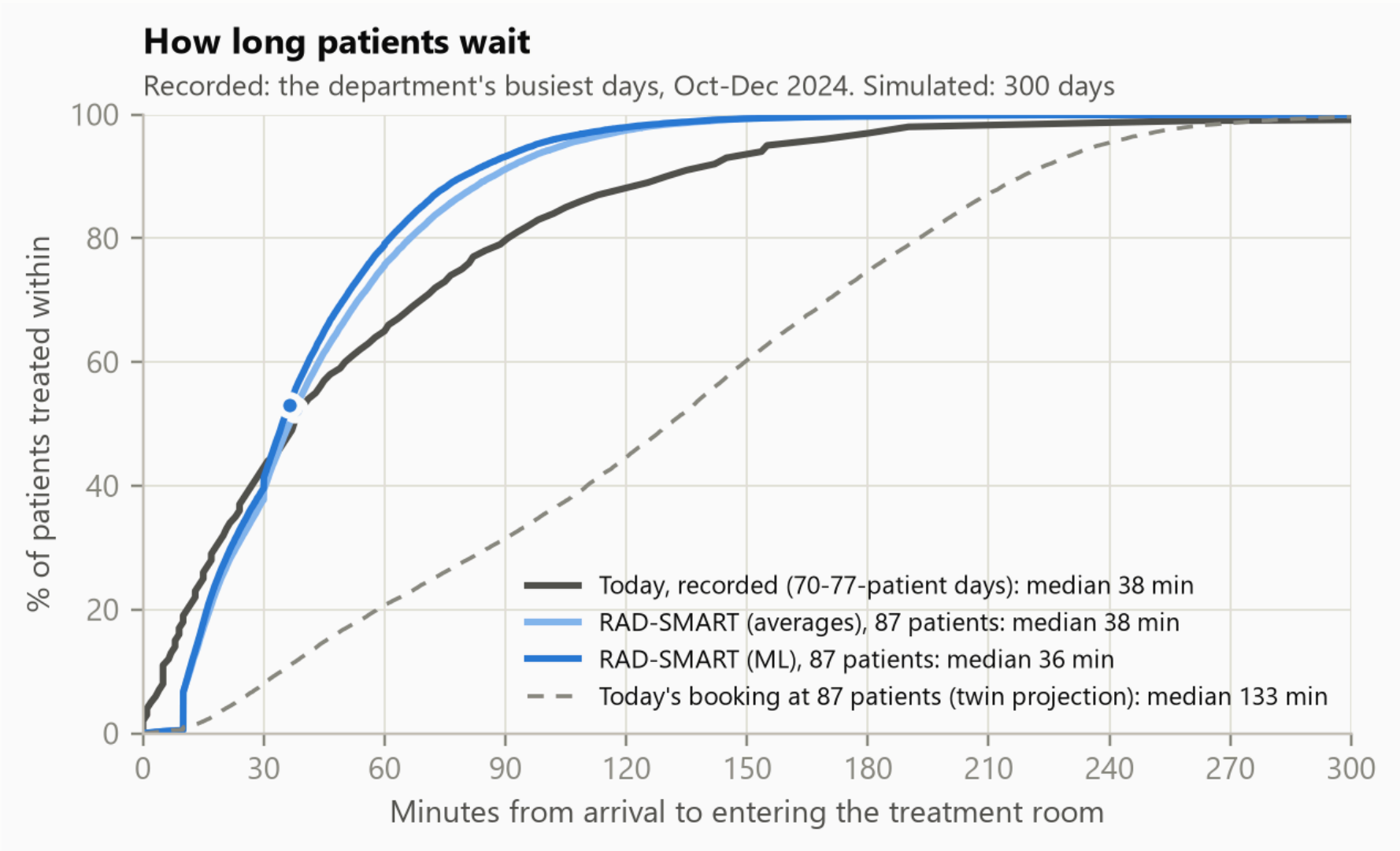
PEOPLE DECIDE

Two days ahead the HoD confirms who starts · the evening before, a senior RTT or oncologist approves the plan · during the day a fault or urgent start triggers a re-plan in seconds, approved before anyone is messaged · overrides and actual times retrain the model weekly

One optimised day, with every rule held



Fewer long waits, and times that hold



130 → 76 min

90th-percentile wait, against the busiest recorded days

27% → 6%

treated more than an hour after their time

66 → 36 min

median wait for evening arrivals: no build-up

104 → 36 min

150-min machine fault: the same plan without → with live re-planning

The typical wait stays about the same (38 → 36 min, preparation included), and 53% are treated within ± 15 min of their time instead of 30%. What goes is the long, unpredictable wait.

Safe by design, then 90 days on the Versa HD

Safe by design

- **Proposal until approved.** A senior RTT or oncologist approves every plan, re-plan and deferral
- **Rules re-checked.** An independent checker verifies all 7 hard rules before publishing
- **Clinicians decide.** No diagnoses, doses, images or notes are read
- **No identifiers to AI.** Models see pseudonymous IDs; names stay on the hospital server
- **Full audit trail.** Every plan, override, rule version and message is logged
- **Synthetic and anonymised only.** Records are used as aggregates; DPDP-aligned

Days 1–14 Baseline

Room entry and exit taps added

Days 15–35 Shadow mode

RAD-SMART plans nightly; RTTs compare, nothing changes

Days 36–90 Live

A senior RTT or oncologist approves each plan; patients get times and live updates; model retrained weekly

PRIMARY KPI

$\geq 25\%$ lower

90th-percentile wait from arrival to linac entry (records: 109 min)

ALSO TRACKED

- Over an hour late: 22% → 10% or fewer
- At least 45% within ± 15 min (today 29%)
- Median wait, overtime and throughput no worse
- Hard rules kept; no group worse off

Next: build sprint 5 Oct – 8 Nov · pilot in Q1 2027 · **our ask:** a pilot slot with mentors, Sarvam API credits, NCG introductions